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Q1. Smart Document Editor using String, StringBuilder and StringBuffer

CODE

```
import java.util.*;
import java.util.regex.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String paragraph = sc.nextLine();
        int characters = paragraph.length();
        String[] words = paragraph.trim().split("\\s+");
        int wordCount = words.length;

        String[] sentences = paragraph.split("[.!?]");
        int sentenceCount = 0;
        for (String s : sentences) {
            if (!s.trim().isEmpty())
                sentenceCount++;
        }
        String oldWord = sc.next();
        String newWord = sc.next();
        String replaced = paragraph.replaceAll("(?i)" + Pattern.quote(oldWord), Matcher.quoteReplacement(newWord));
        StringBuffer history = new StringBuffer();
        history.append("Replaced '")
                .append(oldWord)
                .append("' with '")
                .append(newWord)
                .append("'\n");
        String[] arr = replaced.split("[.!?]");
        StringBuilder finalDoc = new StringBuilder();
        for (String s : arr) {
            s = s.trim();
            if (!s.isEmpty()) {
                StringBuilder temp = new StringBuilder(s);
                finalDoc.append(temp.reverse()).append(". ");
            }
        }
        System.out.println(" Original Document ");
        System.out.println(paragraph);
        System.out.println("\nCharacters : " + characters);
        System.out.println("Words      : " + wordCount);
        System.out.println("Sentences  : " + sentenceCount);
        System.out.println("\n Editing History ");
        System.out.println(history);
        System.out.println(" Final Document ");
        System.out.println(finalDoc);
    }
}
```

INPUT

```
Java is fun.I like java
java
python
```

OUTPUT

```
(no output)
```

Q2. Hospital Token Management System using List Interface

CODE

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<Patient> list = new ArrayList<>();
        int n = sc.nextInt();
        sc.nextLine();
        for (int i = 0; i < n; i++) {
            int id = sc.nextInt();
            sc.nextLine();
            String name = sc.nextLine();
            String dept = sc.nextLine();
            list.add(new Patient(id, name, dept));
        }
        System.out.println("All Patients:");
        for (Patient p : list) {
            System.out.println(p.id + " " + p.name + " " + p.dept);
        }
        int id = sc.nextInt();
        System.out.println("The Entered patien ID to update is : " + id);
        sc.nextLine();
        String newName = sc.nextLine();
        System.out.println("The Entered name to update : " + newName);
        for (Patient p : list) {
            if (p.id == id) {
                p.name = newName;
            }
        }
        String dep = sc.nextLine();
        System.out.println("Enter Department to search : " + dep);
        System.out.println("Patients in " + dep + " Department:");
        for (Patient p : list) {
            if (p.dept.equalsIgnoreCase(dep)) {
                System.out.println(p.id + " " + p.name);
            }
        }
        System.out.println("\nFirst Registered Patient:");
        System.out.println(list.get(0).id + " " + list.get(0).name);
        System.out.println("Last Registered Patient:");
        System.out.println(list.get(list.size() - 1).id + " " + list.get(list.size() - 1).name);
    }
}
class Patient {
    int id;
    String name;
    String dept;
    Patient(int id, String name, String dept) {
        this.id = id;
        this.name = name;
        this.dept = dept;
    }
}
```

INPUT

```
3
101
Ram
Cardiology
102
Ravi
Neurology
103
Kumar
Cardiology
102
```

```
Arun
Cardiology
```

OUTPUT

```
(no output)
```

Q3. Airport Boarding Queue Management using Queue Interface

CODE

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Queue queue = new LinkedList<>();
        PriorityQueue vip = new PriorityQueue<>();
        int n = sc.nextInt();
        sc.nextLine();
        for (int i = 0; i < n; i++) {
            queue.add(sc.nextLine());
        }
        System.out.println("Current Queue: " + queue);
        System.out.println("Next Passenger: " + queue.peek());
        System.out.println("Boarded Passenger: " + queue.poll());
        System.out.println("Remaining Queue: " + queue);
        int m = sc.nextInt();
        sc.nextLine();
        for (int i = 0; i < m; i++) {
            vip.add(sc.nextLine());
        }
        System.out.println("VIP Queue: " + vip);
    }
}
```

INPUT

```
3
Ram
Ravi
Kumar
2
Arun
Bala
```

OUTPUT

```
(no output)
```

Q4. Digital Certificate Verification using Set Interface

CODE

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        LinkedHashSet set = new LinkedHashSet<>();
        TreeSet sorted = new TreeSet<>();
        int n = sc.nextInt();
        for (int i = 0; i < n; i++) {
            int id = sc.nextInt();
            set.add(id);
            sorted.add(id);
        }
        System.out.println("Certificates (Insertion Order): " + set);
        System.out.println("Certificates (Sorted Order): " + sorted);
        System.out.println("Total Unique Certificates: " + set.size());
        int search = sc.nextInt();
        System.out.println("The Entered Certificate ID to search is : " + search);
        if (set.contains(search)) {
            System.out.println("Certificate Found");
        } else {
            System.out.println("Certificate Not Found");
        }
    }
}
```

INPUT

```
5
120
158
102
100
118
100
```

OUTPUT

```
(no output)
```

Q5. Smart Inventory Management using Map Interface

CODE

```
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        HashMap<Integer, Product> map = new HashMap<>();
        int n = sc.nextInt();
        for (int i = 0; i < n; i++) {
            int id = sc.nextInt();
            sc.nextLine();
            String name = sc.nextLine();
            String category = sc.nextLine();
            int quantity = sc.nextInt();
            double price = sc.nextDouble();
            map.put(id, new Product(id, name, category, quantity, price));
        }
        int searchId = sc.nextInt();
        if (map.containsKey(searchId)) {
            Product p = map.get(searchId);
            System.out.println("Product Found");
            System.out.println(p.id + " " + p.name + " " + p.category + " "+ p.quantity + " " + p.price);
        } else {
            System.out.println("Product Not Found");
        }
        double total = 0;
        for (Product p : map.values()) {
            total += p.quantity * p.price;
        }
        System.out.println("Total Inventory Value = " + total);
        sc.nextLine();
        String cat = sc.nextLine();
        for (Product p : map.values()) {
            if (p.category.equalsIgnoreCase(cat)) {
                System.out.println(p.id + " " + p.name + " "+ p.quantity + " " + p.price);
            }
        }
        for (Product p : map.values()) {
            System.out.println(p.id + " " + p.name + " " + p.category + " "+ p.quantity + " " + p.price);
        }
    }
}
class Product {
    int id;
    String name;
    String category;
    int quantity;
    double price;
    Product(int id, String name, String category, int quantity, double price) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.quantity = quantity;
        this.price = price;
    }
}
```

INPUT

```
3
101
Laptop
Electronic
5
50000
102
Mouse
```

```
Electronic
20
500
103
Block
Stationary
15
200
101
Electronics
```

OUTPUT

```
(no output)
```